

S.F. for "Output Feedback Stochastic Model Predictive Control for Discrete-time Stochastic Linear System via Block Coordinate Descent"

APPENDIX A MATRIX DEFINITION

$$\mathcal{A} = \begin{bmatrix} I \\ A \\ A^2 \\ \vdots \\ A^N \end{bmatrix} \in \mathbb{R}^{(N+1)n_x \times n_x},$$

$$\mathcal{K} = \begin{bmatrix} K_0 & 0 & \cdots & \cdots & 0 \\ 0 & K_1 & \ddots & \ddots & \vdots \\ 0 & 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & K_{N-1} & 0 \\ 0 & \cdots & 0 & 0 & 0 \end{bmatrix} \in \mathbb{R}^{(N+1)n_u \times (N+1)n_x}$$

$$\mathcal{B} = \begin{bmatrix} 0 & 0 & \cdots & 0 & 0 \\ B & 0 & \cdots & 0 & 0 \\ AB & B & \ddots & \vdots & 0 \\ \vdots & \ddots & \ddots & 0 & 0 \\ A^{N-1}B & A^{N-2}B & \cdots & B & 0 \end{bmatrix} \in \mathbb{R}^{(N+1)n_x \times (N+1)n_u},$$

$$\mathcal{D} = \begin{bmatrix} 0 & 0 & \cdots & 0 & 0 \\ D & 0 & \cdots & 0 & 0 \\ AD & D & \ddots & \vdots & 0 \\ \vdots & \vdots & \ddots & 0 & 0 \\ A^{N-1}D & A^{N-2}D & \cdots & D & 0 \end{bmatrix} \in \mathbb{R}^{(N+1)n_x \times (N+1)n_w},$$

$$\mathcal{C} = \begin{bmatrix} C & 0 & \cdots & 0 \\ 0 & C & \ddots & \vdots \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & \cdots & C \end{bmatrix} \in \mathbb{R}^{(N+1)n_y \times (N+1)n_x}.$$

$$\mathcal{L} = \begin{bmatrix} 0 & 0 & \cdots & 0 & 0 \\ L_0 & 0 & \cdots & 0 & 0 \\ AL_0 & L_1 & \ddots & \vdots & 0 \\ \vdots & \ddots & \ddots & 0 & 0 \\ A^{N-1}L_0 & A^{N-2}L_1 & \cdots & L_{N-1} & 0 \end{bmatrix} \in \mathbb{R}^{(N+1)n_x \times (N+1)n_x}.$$

Furthermore, we introduce the following augmented state vectors:

$$\begin{aligned} X &= [x_0^T \ x_1^T \ x_2^T \ \cdots \ x_N^T]^T \in \mathbb{R}^{(N+1)n_x}, \\ \hat{X} &= [\hat{x}_0^T \ \hat{x}_1^T \ \hat{x}_2^T \ \cdots \ \hat{x}_N^T]^T \in \mathbb{R}^{(N+1)n_x}, \\ \bar{X} &= [\bar{x}_0^T \ \bar{x}_1^T \ \bar{x}_2^T \ \cdots \ \bar{x}_N^T]^T \in \mathbb{R}^{(N+1)n_x}, \\ U &= [u_0^T \ u_1^T \ \cdots \ u_{N-1}^T \ 1_{n_u}^T]^T \in \mathbb{R}^{(N+1)n_u}, \\ W(\bullet) &= [w_{(\bullet)0}^T \ w_{(\bullet)1}^T \ \cdots \ w_{(\bullet)N-1}^T \ 1_{n_w}^T]^T \in \mathbb{R}^{(N+1)n_w}, \\ V &= [v_0^T \ v_1^T \ \cdots \ v_{N-1}^T \ 1_{n_u}^T]^T \in \mathbb{R}^{(N+1)n_u}, \end{aligned}$$

$$\begin{aligned} E &= [e_0^T \ e_1^T \ \cdots \ e_N^T]^T \in \mathbb{R}^{(N+1)n_x}, \\ F &= [f_0^T \ f_1^T \ \cdots \ f_N^T]^T \in \mathbb{R}^{(N+1)n_x}. \end{aligned}$$

$$\mathcal{V} = \begin{cases} V & \left\{ \begin{aligned} \bar{X} &= \mathcal{A}\bar{x}_0 + \mathcal{B}V \\ \Sigma_E &= [\mathbf{I} - \tilde{\mathcal{L}}^k \ \tilde{\mathcal{L}}^k \mathcal{C}^+] \\ &\times \begin{bmatrix} z_e & 0 \\ 0 & \Sigma_{W_2} \end{bmatrix} \begin{bmatrix} (\mathbf{I} - \tilde{\mathcal{L}}^k) \\ (\tilde{\mathcal{L}}^k \mathcal{C}^+)^T \end{bmatrix} \\ \Sigma_F &= (\mathbf{I} + \mathcal{B}\tilde{\mathcal{K}}^k) \\ &\times [\mathbf{I} \ \mathbf{I}] \begin{bmatrix} z_f & 0 \\ 0 & \Sigma_E \end{bmatrix} \begin{bmatrix} \mathbf{I} \\ \mathbf{I} \end{bmatrix} (\mathbf{I} + \mathcal{B}\tilde{\mathcal{K}}^k)^T \\ a_{x,j}^T M_{t,x} \bar{X} + \Psi_{1-\Delta_x}^{-1} \\ &\times \sqrt{a_{x,j}^T M_{t,x} 2(\Sigma_E + \Sigma_F) (a_{x,j}^T M_{t,x})^T} \leq b_{x,j} \\ a_{u,l}^T M_{t,u} V + \Psi_{1-\Delta_u}^{-1} \\ &\times \sqrt{a_{u,l}^T M_{t,u} \tilde{\mathcal{K}}^k \Sigma_F (\tilde{\mathcal{K}}^k)^T (a_{u,l}^T M_{t,u})^T} \leq b_{u,l} \end{aligned} \right\} \\ \tilde{\mathcal{L}} & \left\{ \begin{aligned} \bar{X} &= \mathcal{A}\bar{x}_0 + \mathcal{B}V^{k+1} \\ \Sigma_E &= [\mathbf{I} - \tilde{\mathcal{L}} \ \tilde{\mathcal{L}} \mathcal{C}^+] \\ &\times \begin{bmatrix} z_e & 0 \\ 0 & \Sigma_{W_2} \end{bmatrix} [\mathbf{I} - \tilde{\mathcal{L}} \ \tilde{\mathcal{L}} \mathcal{C}^+]^T \\ \Sigma_F &= (\mathbf{I} + \mathcal{B}\tilde{\mathcal{K}}^k) \\ &\times [\mathbf{I} \ \mathbf{I}] \begin{bmatrix} z_f & 0 \\ 0 & \Sigma_E \end{bmatrix} \begin{bmatrix} \mathbf{I} \\ \mathbf{I} \end{bmatrix} (\mathbf{I} + \mathcal{B}\tilde{\mathcal{K}}^k)^T \\ a_{x,j}^T M_{t,x} \bar{X} + \Psi_{1-\Delta_x}^{-1} \\ &\times \sqrt{a_{x,j}^T M_{t,x} 2(\Sigma_E + \Sigma_F) (a_{x,j}^T M_{t,x})^T} \leq b_{x,j} \\ a_{u,l}^T M_{t,u} V^{k+1} + \Psi_{1-\Delta_u}^{-1} \\ &\times \sqrt{a_{u,l}^T M_{t,u} \Sigma_{F,K} (a_{u,l}^T M_{t,u})^T} \leq b_{u,l} \end{aligned} \right\} \end{cases}$$

$$\mathcal{Z} = \left\{ \tilde{\mathcal{K}} \left| \begin{array}{l} \bar{X} = \mathcal{A}\bar{x}_0 + \mathcal{B}V^{k+1} \\ \Sigma_E = [\mathbf{I} - \tilde{\mathcal{L}}^{k+1} \quad \tilde{\mathcal{L}}^{k+1}\mathcal{C}^+] \\ \quad \times \begin{bmatrix} z_e & 0 \\ 0 & \Sigma_{W_2} \end{bmatrix} \begin{bmatrix} (\mathbf{I} - \tilde{\mathcal{L}}^{k+1})^T \\ (\tilde{\mathcal{L}}^{k+1}\mathcal{C}^+)^T \end{bmatrix} \\ \Sigma_F = (\mathbf{I} + \mathcal{B}\tilde{\mathcal{K}}) \\ \quad \times [\mathbf{I} \quad \mathbf{I}] \begin{bmatrix} z_f & 0 \\ 0 & \Sigma_E \end{bmatrix} \begin{bmatrix} \mathbf{I} \\ \mathbf{I} \end{bmatrix} (\mathbf{I} + \mathcal{B}\tilde{\mathcal{K}})^T \\ a_{x,j}^T M_{t,x} \bar{X} + \Psi_{1-\Delta_x}^{-1} \\ \quad \times \sqrt{a_{x,j}^T M_{t,x} 2(\Sigma_E + \Sigma_F) (a_{x,j}^T M_{t,x})^T} \leq b_{x,j} \\ a_{u,l}^T M_{t,u} V^{k+1} + \Psi_{1-\Delta_u}^{-1} \\ \quad \times \sqrt{a_{u,l}^T M_{t,u} \Sigma_{F,K} (a_{u,l}^T M_{t,u})^T} \leq b_{u,l} \end{array} \right\}$$

for $t = 0, \dots, N-1$, $j = 1, \dots, m_x$ and $l = 1, \dots, m_u$.

APPENDIX B PROOF OF THEOREM 1

Before we proof the Theorem 1, we first provide the following results.

Lemma 2: Let $X \sim \mathcal{N}(\mu_X, \Sigma_X)$, $E \sim \mathcal{N}(0, \Sigma_E)$ and $F \sim \mathcal{N}(0, \Sigma_F)$ be random variables on $\mathbb{R}$. If they satisfy (37) and (11), then $\mu_X = \bar{X}$ and $\Sigma_X \preceq 2(\Sigma_E + \Sigma_F)$.

Proof: Since $\mu_X = \bar{X}$ is obvious, its proof is omitted.

Define $\Phi = \begin{bmatrix} E & 0 \\ 0 & F \end{bmatrix}$. Since $X = E + F + \bar{X}$, we have

$$\Sigma_X = [\mathbf{I}, \mathbf{I}] \Sigma_\Phi [\mathbf{I}, \mathbf{I}]^T$$

Furthermore, we have

$$\Sigma_\Phi = \begin{bmatrix} \Sigma_E & \text{cov}(E, F) \\ \text{cov}(E, F)^T & \Sigma_F \end{bmatrix}$$

Since for all $M \succcurlyeq 0$ and all $\begin{bmatrix} z_1^T & z_2^T \end{bmatrix} \neq 0$, we have

$$\begin{aligned} & \begin{bmatrix} -z_1^T & z_2^T \end{bmatrix} \begin{bmatrix} M_{11} & M_{12} \\ M_{12}^T & M_{22} \end{bmatrix} \begin{bmatrix} -z_1 \\ z_2 \end{bmatrix} \\ &= z_1^T M_{11} z_1 + z_2^T M_{22} z_2 \\ &\quad - z_1^T M_{12} z_2 - z_2^T M_{12}^T z_1 \succcurlyeq 0, \end{aligned}$$

Thus, we can obtain that

$$\begin{aligned} & \begin{bmatrix} z_1^T & z_2^T \end{bmatrix} \begin{bmatrix} M_{11} & M_{12} \\ M_{12}^T & M_{22} \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} \\ &= z_1^T M_{11} z_1 + z_2^T M_{22} z_2 + z_1^T M_{12} z_2 + z_2^T M_{12}^T z_1 \\ &\leq 2z_1^T M_{11} z_1 + 2z_2^T M_{22} z_2 \\ &= \begin{bmatrix} z_1^T & z_2^T \end{bmatrix} \begin{bmatrix} 2M_{11} & 0 \\ 0 & 2M_{22} \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} \end{aligned}$$

Since $\Sigma_\Phi \succcurlyeq 0$, we have

$$\begin{bmatrix} \Sigma_E & \text{cov}(E, F) \\ \text{cov}(E, F)^T & \Sigma_F \end{bmatrix} \preceq \begin{bmatrix} 2\Sigma_E & 0 \\ 0 & 2\Sigma_F \end{bmatrix}$$

This implies $\Sigma_X = [\mathbf{I}, \mathbf{I}] \Sigma_\Phi [\mathbf{I}, \mathbf{I}]^T \preceq 2\Sigma_E + 2\Sigma_F$. This completes the proof. ■

Based on Lemma 2, we have

$$\begin{aligned} M_{0,x} \mu_X &= M_{0,x} \bar{X} \\ M_{0,x} \Sigma_X M_{0,x}^T &\preceq M_{0,x} (2(\Sigma_E + \Sigma_F)) M_{0,x}^T \end{aligned}$$

and

$$\begin{aligned} M_{N,x} \mu_X &= M_{N,x} \bar{X} \\ M_{N,x} \Sigma_X M_{N,x}^T &\preceq M_{N,x} (2(\Sigma_E + \Sigma_F)) M_{N,x}^T \end{aligned}$$

Therefore, we can derive the results directly.

APPENDIX C PROOF OF LEMMA 1

Proof. The objective function $\tilde{J}(V, \tilde{\mathcal{L}}, \tilde{\mathcal{K}})$ is continuously differentiable and bounded below by zero (since $\mathcal{Q} \geq 0$ and $\mathcal{R} \geq 0$). In Algorithm 1, the sub-problems corresponding to V , $\tilde{\mathcal{L}}$, and $\tilde{\mathcal{K}}$ are rigorously converted into a convex Quadratic Program (QP), a Second-Order Cone Program (SOCP), and another SOCP, respectively. Thus, each sub-problem is solved to global optimality for its respective block while fixing the other variables. We therefore obtain:

$$\begin{aligned} \tilde{J}(V^{k+1}, \tilde{\mathcal{L}}^{k+1}, \tilde{\mathcal{K}}^{k+1}) &\leq \tilde{J}(V^{k+1}, \tilde{\mathcal{L}}^{k+1}, \tilde{\mathcal{K}}^k) \\ &\leq \tilde{J}(V^{k+1}, \tilde{\mathcal{L}}^k, \tilde{\mathcal{K}}^k) \\ &\leq \tilde{J}(V^k, \tilde{\mathcal{L}}^k, \tilde{\mathcal{K}}^k) \end{aligned}$$

Since the sequence of objective values is non-increasing and bounded below, it converges to a finite infimum. Under the strict convexity of the individual QP and SOCP sub-problems, the minimizer for each block is uniquely attained. According to the convergence theory for block coordinate descent methods for continuously differentiable functions, every limit point of the sequence is a stationary point [41]. This completes the proof. □

APPENDIX D PROOF OF THEOREM 2

Proof. Assume that at time $t-1$, Problem 3 is feasible and yields the optimal solution sequence V_{t-1}^* , $\tilde{\mathcal{L}}_{t-1}^*$, and $\tilde{\mathcal{K}}_{t-1}^*$ (as guaranteed by Lemma 1). At time t , the controller selects the initial condition according to (80).

Case 1: Problem 3 is feasible with the current estimation $(\hat{\mu}_{x,t}, \hat{\Sigma}_{x,t})$. The initialization strategy directly accepts it, i.e., $(\mu_{x,0|t}, \Sigma_{x,0|t}) = (\hat{\mu}_{x,t}, \hat{\Sigma}_{x,t})$. A feasible solution explicitly exists, rendering the problem trivially feasible.

Case 2: Problem 3 is infeasible with the current estimation. The strategy falls back to the deterministic surrogate bounds optimized from the previous step: $(\mu_{x,0|t}, \Sigma_{x,0|t}) = (\bar{x}_{1|t-1}^*, 2(\Sigma_{E,1|t-1}^* + \Sigma_{F,1|t-1}^*))$. We construct a candidate shifted sequence by appending the terminal gains K_f , L_f , and a nominal terminal input $v_f = 0$:

$$\begin{aligned} V_t^{cand} &= [v_{1|t-1}^*, \dots, v_{N-1|t-1}^*, 0] \\ K_t^{cand} &= [K_{1|t-1}^*, \dots, K_{N-1|t-1}^*, K_f] \\ L_t^{cand} &= [L_{1|t-1}^*, \dots, L_{N-1|t-1}^*, L_f] \end{aligned}$$

By initializing strictly with the surrogate upper bound optimized at $t-1$, the nominal states and surrogate covariance matrices for the first $N-1$ steps perfectly match

the optimal tail of the previous iteration. Specifically, for all $i \in \{0, \dots, N-1\}$, we have the exact index-shifted propagation:

$$\bar{x}_{i|t}^{cand} = \bar{x}_{i+1|t-1}^*, \Sigma_{E,i|t}^{cand} = \Sigma_{E,i+1|t-1}^*, \Sigma_{F,i|t}^{cand} = \Sigma_{F,i+1|t-1}^*$$

Substituting these shifted variables into the tightened surrogate state chance constraints at time t yields:

$$\begin{aligned} & a_{x,j}^T \bar{x}_{i|t}^{cand} + \Psi_{1-\Delta_x}^{-1} \sqrt{2a_{x,j}^T (\Sigma_{E,i|t}^{cand} + \Sigma_{F,i|t}^{cand}) a_{x,j}} \\ &= a_{x,j}^T \bar{x}_{i+1|t-1}^* + \\ & \Psi_{1-\Delta_x}^{-1} \sqrt{2a_{x,j}^T (\Sigma_{E,i+1|t-1}^* + \Sigma_{F,i+1|t-1}^*) a_{x,j}} \\ & \leq b_{x,j} \end{aligned}$$

which are strictly satisfied since the optimal sequence at $t-1$ was strictly feasible. By the property established in Lemma 2 ($\Sigma_X \preccurlyeq 2(\Sigma_E + \Sigma_F)$), satisfying these surrogate constraints rigorously guarantees the satisfaction of the actual physical chance constraints:

$$a_{x,j}^T \mu_{X,i|t} + \Psi_{1-\Delta_x}^{-1} \sqrt{a_{x,j}^T \Sigma_{X,i|t} a_{x,j}} \leq b_{x,j}$$

Similar index-shifted inequality propagation holds for the input chance constraints and the terminal covariance constraint $M_N(\Sigma_X) M_N^T \preccurlyeq \Sigma_{x,N}$. For the final prediction step N , the shifted terminal state enters the terminal invariant set Ω_f . Under Assumption 1, the local laws ensure all probabilistic terminal constraints and covariance bounds are met. Therefore, the constructed shifted sequences constitute a strictly feasible solution. By mathematical induction, the system maintains recursive feasibility. $\square$

APPENDIX E PROOF OF THEOREM 3

Proof. Since $\mathcal{Q} \geq 0$ and $\mathcal{R} \geq 0$, we have that $l_{i|t}^m \geq 0$. Then $J^m(\mu_{X,t}, V_t)$ is positive. From the recursive feasibility established previously, $v_{i|t} \in \mathcal{U}$ is bounded and so is $\bar{x}_{i|t} \in \mathcal{X}$. Then $J^m(\mu_{X,t}, V_t)$ is bounded.

When the system utilizes Strategy 2, the initialization strictly follows the prediction, and we have the standard deterministic decrease:

$$\begin{aligned} & J^m(\mu_{X,t}, V_t) - J^m(\mu_{X,t-1}, V_{t-1}) \\ & \leq v_{N|t}^T R v_{N|t} + \bar{x}_{N+1|t-1}^{*T} Q \bar{x}_{N+1|t-1}^* \\ & \quad - v_{0|t-1}^{*T} R v_{0|t-1}^* - \bar{x}_{0|t}^{*T} Q \bar{x}_{0|t}^* \leq 0 \end{aligned}$$

When Strategy 1 is triggered, the initial nominal state incorporates the latest measurement estimate $\hat{\mu}_{x,t}$. The deviation between the measurement estimate and the prior prediction $\bar{x}_{1|t-1}^*$ is governed by the bounded estimation error dynamics characterized by Σ_E . Because the objective function is locally Lipschitz continuous within the compact feasible sets $\mathcal{X}$ and $\mathcal{U}$, there exists a finite positive constant C_w bounded by the noise covariances Σ_{w_1} and Σ_{w_2} such that the cost variation induced by the measurement update is bounded:

$$J^m(\hat{\mu}_{X,t}, V_t) \leq J^m(\bar{x}_{1|t-1}^*, V_{t-1}) + C_w$$

Therefore, the expected value of the Lyapunov function strictly decreases outside a bounded set defined by C_w , establishing that the nominal closed-loop system is practically stable and converges to a bounded region around the origin rather than exactly to zero. $\square$

APPENDIX F PROOF OF THEOREM 4

Proof. Let the optimal cost at time t be $J_t^* = J(\Sigma_{X,t}^*, \mu_{X,t}^*, K_t^*, V_t^*)$. At time $t+1$, under recursive feasibility, the shifted sequence $(V_{t+1}^{cand}, K_{t+1}^{cand}, L_{t+1}^{cand})$ with $v_f = 0$ and terminal gains K_f, L_f constitutes a valid feasible solution. Let the cost associated with this candidate sequence be J_{t+1}^{cand} . By optimality, we have:

$$J_{t+1}^* \leq J_{t+1}^{cand} \quad (1)$$

The candidate cost J_{t+1}^{cand} at time $t+1$ is constructed by removing the first stage cost of time t and adding the terminal stage cost. Before addressing the closed-loop measurement update, we first explicitly derive the algebraic upper bound for the terminal stage cost $l_{N+1|t}^{cand} = l(\bar{x}_{N+1|t}^*, v_f, \Sigma_{x_{N+1|t}}^*, K_f)$.

From the cost definition, the terminal stage cost consists of nominal and variance components:

$$\begin{aligned} l_{N+1|t}^{cand} &= \bar{x}_{N+1|t}^{*T} Q \bar{x}_{N+1|t}^* + v_f^T R v_f \\ & \quad + \text{Tr}(\Sigma_{x_{N+1|t}}^* Q) + \text{Tr}(K_f \Sigma_{F,N+1|t}^* K_f^T R) \end{aligned}$$

Since $v_f \in \mathcal{U}$ and $\bar{x}_{N+1|t}^* \in \mathcal{X}$ within the terminal invariant set, their nominal costs are bounded by finite maximums: $v_f^T R v_f \leq J_v^{max}$ and $\bar{x}_{N+1|t}^{*T} Q \bar{x}_{N+1|t}^* \leq J_x^{max}$.

From the surrogate terminal constraint and Lemma 2, we have $\Sigma_{x_{N+1|t}}^* \preccurlyeq 2(\Sigma_E + \Sigma_F) \preccurlyeq \Sigma_{x,N}$. Thus, $\text{Tr}(\Sigma_{x_{N+1|t}}^* Q) \leq \text{Tr}(\Sigma_{x,N} Q)$. From the input chance constraint reformulation, we know:

$$\begin{aligned} & \Psi_{1-\Delta_u}^{-1} \sqrt{a_u^T K_f \Sigma_F K_f^T a_u} \leq b_u \\ \implies & K_f \Sigma_F K_f^T \preccurlyeq a_u^{-T} \left(\frac{b_u}{\Psi_{1-\Delta_u}^{-1}} \right)^2 a_u^{-1} \end{aligned}$$

Consequently, the terminal feedback variance cost is bounded by $\text{Tr} \left(a_u^{-T} \left(\frac{b_u}{\Psi_{1-\Delta_u}^{-1}} \right)^2 a_u^{-1} R \right)$. Defining the deterministic terminal upper bound ξ as:

$$\xi = J_v^{max} + J_x^{max} + \text{Tr}(\Sigma_{x,N} Q) + \text{Tr} \left(a_u^{-T} \left(\frac{b_u}{\Psi_{1-\Delta_u}^{-1}} \right)^2 a_u^{-1} R \right) \quad (2)$$

we algebraically guarantee that $l_{N+1|t}^{cand} \leq \xi$.

Unlike a nominal deterministic case, the actual realization of the candidate cost at $t+1$ depends on the closed-loop initialization strategy. If Strategy 1 is triggered, the state undergoes a measurement update, introducing an innovation jump defined as $\delta_t = \hat{x}_t - \bar{x}_{1|t-1}^*$.

Since the actual control input applied is $u_{t-1} = v_{0|t-1}^* + K_{0|t-1}^*(\hat{x}_{t-1} - \bar{x}_{0|t-1}^*)$, and Strategy 1 implies $\bar{x}_{0|t-1}^* = \hat{x}_{t-1}$,

the feedback term vanishes, yielding $u_{t-1} = v_{0|t-1}^*$. Substituting the measurement model $y_{t-1} = Cx_{t-1} + w_{2,t-1}$ into the observer dynamics, the nominal terms perfectly cancel out, simplifying the innovation jump to:

$$\delta_t = L_{t-1}^* (Ce_{0|t-1} + w_{2,t-1}) \quad (3)$$

Given that $e_{0|t-1}$ and $w_{2,t-1}$ are zero-mean and mutually independent, it follows that $\mathbb{E}[\delta_t] = 0$. The covariance of this jump, denoted as Σ_δ , is strictly determined by the observer gain and noise statistics:

$$\Sigma_\delta = \mathbb{E}[\delta_t \delta_t^T] = L_{t-1}^* \left(C \Sigma_{E,0|t-1}^* C^T + \Sigma_{w_2} \right) L_{t-1}^{*T} \quad (4)$$

As the candidate control sequence V_{t+1}^{cand} is open-loop shifted, this zero-mean jump δ_t propagates linearly through the prediction horizon as $A^i \delta_t$. Evaluating its expectation within the nominal quadratic cost J^m , the cross-terms vanish ($\mathbb{E}[\delta_t] = 0$), yielding an explicit closed-form bound Γ for the expected additional innovation cost:

$$\Gamma = \sum_{i=0}^N \text{Tr} \left((A^i)^T Q A^i \Sigma_\delta \right) \quad (5)$$

This explicit mathematical formulation is of paramount importance: it not only proves that Γ is a finite bound (restricted by noise variances and the prediction horizon), but also rigorously justifies the necessity of the proposed BCD algorithm. By jointly optimizing the observer gain L_t iteratively, the algorithm structurally minimizes the quadratic term Σ_δ , thereby systematically suppressing the innovation cost Γ .

Taking the conditional expectation given the information at time t , the expected cost difference is upper bounded by the terminal cost addition (ξ), the initial stage cost reduction, and the explicit innovation bound Γ :

$$\begin{aligned} \mathbb{E}_w[J_{t+1}^{cand}] - J_t^* &\leq l_{N+1|t}^{cand} - l(x_{0|t}^*, u_{0|t}^*) + \Gamma \\ &\leq \xi + \Gamma - l(x_{0|t}^*, u_{0|t}^*) \end{aligned} \quad (6)$$

Defining the unified closed-loop bound $\bar{\xi} = \xi + \Gamma$, and utilizing the optimality condition $J_{t+1}^* \leq J_{t+1}^{cand}$, we obtain:

$$\mathbb{E}_w[J_{t+1}^*] - J_t^* \leq \bar{\xi} - l(x_{0|t}^*, u_{0|t}^*) \quad (7)$$

Taking the total expectation and summing this inequality from $t = 0$ to N yields a telescoping sum without requiring the open-loop assumption $v_{i+1|t}^* = v_{i|t+1}^*$:

$$\sum_{t=0}^N \mathbb{E}_w(l(x_{0|t}^*, u_{0|t}^*)) \leq J_0^* - \mathbb{E}_w(J_{N+1}^*) + (N+1)\bar{\xi} \quad (8)$$

Since $J_{N+1}^* \geq 0$, we drop the $-\mathbb{E}_w(J_{N+1}^*)$ term. Dividing both sides by N and taking the limit as $N \rightarrow \infty$, we obtain:

$$\lim_{N \rightarrow \infty} \frac{1}{N} \mathbb{E}_w \left(\sum_{t=0}^N l(x_{0|t}^*, u_{0|t}^*) \right) \leq \bar{\xi} \quad (9)$$

This rigorously proves that the expected average stage cost is strictly bounded for the closed-loop stochastic system, explicitly capturing both terminal constraint bounds and measurement feedback innovations. $\square$